## Chpater1 Mechanics & Fluids

## Lesson1 Linear Motion

**1.** The position versus time for a certain particle moving along the x axis is shown in Figure P2.1. Find the average velocity in the time intervals (a) 0 to 2 s, (b) 0 to 4 s, (c) 2 s to 4 s, (d) 4 s to 7 s, and (e) 0 to 8 s.

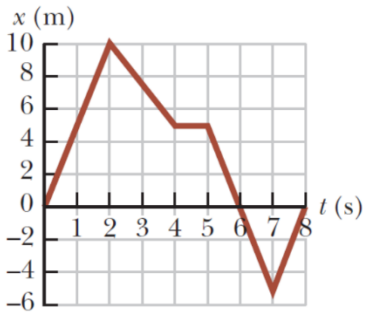



Figure P2.1 Problems 1 and 9.

**2.** The speed of a nerve impulse in the human body is about 100 m/s. If you accidentally stub your toe in the dark, estimate the time it takes the nerve impulse to travel to your brain.

**3.** A person walks first at a constant speed of 5.00 m/s along a straight line from point A to point B and then back along the line from B to A at a constant speed of 3.00 m/s. **(a)** What is her average speed over the entire trip? **(b)** What is her average velocity over the entire trip?

**4.** A particle moves according to the equation $x = 10t^2$, where $x$ is in meters and t is in seconds. **(a)** Find the average velocity for the time interval from 2.00 s to 3.00 s. **(b)** Find the average velocity for the time interval from 2.00 to 2.10 s.

**5.** The position of a pinewood derby car was observed at various times; the results are summarized in the following table. Find the average velocity of the car for **(a)** the first second, **(b)** the last 3 s, and **(c)** the entire period of observation.

| ***t* (s)** | 0 | 1.0 | 2.0 | 3.0 | 4.0 | 5.0 |
|---|---|---|---|---|---|---|
| ***x* (m)** | 0 | 2.3 | 9.2 | 20.7 | 36.8 | 57.5 |

**6.** The position of a particle moving along the $x$ axis varies in time according to the expression $x = 3t^2$, where $x$ is in meters and $t$ is in seconds. Evaluate its position **(a)** at $t = 3.00$ s and **(b)** at 3.00 s + Δt. **(c)** Evaluate the limit of $\Delta x/\Delta t$ as $\Delta t$ approaches zero to find the velocity at $t = 3.00$ s.

**7.** A position–time graph for a particle moving along the $x$ axis is shown in Figure P2.7. **(a)** Find the average velocity in the time interval $t = 1.50$ s to $t = 4.00$ s. **(b)** Determine the instantaneous velocity at $t = 2.00$ s by measuring the slope of the tangent line shown in the graph. **(c)** At what value of t is the velocity zero?

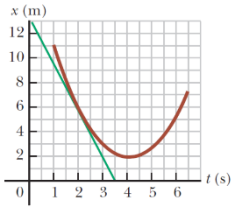



Figure P2.7

**8.** An athlete leaves one end of a pool of length $L$ at $t = 0$ and arrives at the other end at time $t_1$. She swims back and arrives at the starting position at time $t_2$. If she is swimming initially in the positive $x$ direction, determine her average velocities symbolically in **(a)** the first half of the swim, **(b)** the second half of the swim, and **(c)** the round trip. **(d)** What is her average speed for the round trip?

**9.** Find the instantaneous velocity of the particle described in **Figure P2.1** at the following times: **(a)** $t = 1.0$ s, **(b)** $t = 3.0$ s, **(c)** $t = 4.5$ s, and **(d)** $t = 7.5$ s.

**10.** A 50.0-g Super Ball traveling at 25.0 m/s bounces off a brick wall and rebounds at 22.0 m/s. A high-speed camera records this event. If the ball is in contact with the wall for 3.50 ms, what is the magnitude of the average acceleration of the ball during this time interval?

**11.** A velocity–time graph for an object moving along the $x$ axis is shown in **Figure P2.15.** (a) Plot a graph of the acceleration versus time. Determine the average acceleration of the object (b) in the time interval $t = 5.00$ s to $t = 15.0$ s and **(c)** in the time interval $t = 0$ to $t = 20.0$ s.

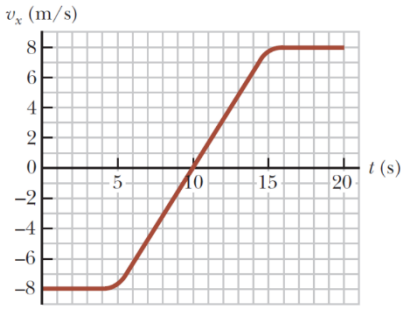



Figure P2.15

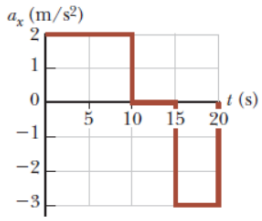



Figure P2.19

**12.** A particle starts from rest and accelerates as shown in **Figure P2.19**. Determine **(a)** the particle's speed at $t = 10.0$ s and at $t = 20.0$ s, and **(b)** the distance traveled in the first 20.0 s.

**13.** An object moves along the x axis according to the equation $x = 3.00t^2 - 2.00t + 3.00$, where $x$ is in meters and $t$ is in seconds. Determine **(a)** the average speed between $t = 2.00$ s and $t = 3.00$ s, **(b)** the instantaneous speed at $t = 2.00$ s and at $t = 3.00$ s, **(c)** the average acceleration between $t = 2.00$ s and $t = 3.00$ s, and **(d)** the instantaneous acceleration at $t = 2.00$ s and $t = 3.00$ s. **(e)** At what time is the object at rest?

**14.** A particle moves along the $x$ axis according to the equation $x = 2.00 + 3.00t - 1.00t^2$, where $x$ is in meters and t is in seconds. At $t = 3.00$ s, find **(a)** the position of the particle, **(b)** its velocity, and **(c)** its acceleration.

**15.** The minimum distance required to stop a car moving at 35.0 mi/h is 40.0 ft. What is the minimum stopping distance for the same car moving at 70.0 mi/h, assuming the same rate of acceleration?

**16.** An electron in a cathode-ray tube accelerates uniformly from $2.00 \times 10^4$ m/s to $6.00 \times 10^6$ m/s over 1.50 cm. **(a)** In what time interval does the electron travel this 1.50 cm? **(b)** What is its acceleration?

**17.** A speedboat moving at $30.0\ \mathrm{m/s}$ approaches a no-wake buoy marker $100\ \mathrm{m}$ ahead. The pilot slows the boat with a constant acceleration of $23.50\ \mathrm{m/s^2}$ by reducing the throttle. **(a)** How long does it take the boat to reach the buoy? **(b)** What is the velocity of the boat when it reaches the buoy?

**18.** A truck covers $40.0\ \mathrm{m}$ in $8.50\ \mathrm{s}$ while smoothly slowing down to a final speed of $2.80\ \mathrm{m/s}$. **(a)** Find its original speed. **(b)** Find its acceleration.

**19.** An object moving with uniform acceleration has a velocity of $12.0\ \mathrm{cm/s}$ in the positive $x$ direction when its $x$ coordinate is $3.00\ \mathrm{cm}$. If its $x$ coordinate $2.00\ \mathrm{s}$ later is $25.00\ \mathrm{cm}$, what is its acceleration?

**20.** A truck on a straight road starts from rest, accelerating at $2.00\ \mathrm{m/s^2}$ until it reaches a speed of $20.0\ \mathrm{m/s}$. Then the truck travels for $20.0\ \mathrm{s}$ at constant speed until the brakes are applied, stopping the truck in a uniform manner in an additional $5.00\ \mathrm{s}$. **(a)** How long is the truck in motion? **(b)** What is the average velocity of the truck for the motion described?

**21.** A particle moves along the x axis. Its position is given by the equation $x = 2 + 3t - 4t^2$, with $x$ in meters and $t$ in seconds. Determine **(a)** its position when it changes direction and **(b)** its velocity when it returns to the position it had at $t = 0$.

**22.** An object moves with constant acceleration $4.00\ \mathrm{m/s^2}$ and over a time interval reaches a final velocity of $12.0\ \mathrm{m/s}$. **(a)** If its initial velocity is $6.00\ \mathrm{m/s}$, what is its displacement during the time interval? **(b)** What is the distance it travels during this interval? **(c)** If its initial velocity is $26.00\ \mathrm{m/s}$, what is its displacement during the time interval? **(d)** What is the total distance it travels during the interval in part **(c)**?

**23.** A hockey player is standing on his skates on a frozen pond when an opposing player, moving with a uniform speed of $12.0\ \mathrm{m/s}$, skates by with the puck. After $3.00\ \mathrm{s}$, the first player makes up his mind to chase his opponent. If he accelerates uniformly at $4.00\ \mathrm{m/s^2}$, **(a)** how long does it take him to catch his opponent and **(b)** how far has he traveled in that time? (Assume the player with the puck remains in motion at constant speed.)

**24.** A baseball is hit so that it travels straight upward after being struck by the bat. A fan observes that it takes $3.00\ \mathrm{s}$ for the ball to reach its maximum height. Find **(a)** the ball's initial velocity and **(b)** the height it reaches.

**25.** The height of a helicopter above the ground is given by $h = 3.00t^3$, where $h$ is in meters and $t$ is in seconds. At $t = 2.00\ \mathrm{s}$, the helicopter releases a small mailbag. How long after its release does the mailbag reach the ground?

**26.** A ball is thrown directly downward with an initial speed of $8.00\ \mathrm{m/s}$ from a height of $30.0\ \mathrm{m}$. After what time interval does it strike the ground?

**27.** A ball is thrown upward from the ground with an initial speed of $25\ \mathrm{m/s}$; at the same instant, another ball is dropped from a building $15\ \mathrm{m}$ high. After how long will the balls be at the same height above the ground?

**28.** A student throws a set of keys vertically upward to her sorority sister, who is in a window $4.00\ \mathrm{m}$ above. The second student catches the keys $1.50\ \mathrm{s}$ later. **(a)** With what initial velocity were the keys thrown? **(b)** What was the velocity of the keys just before they were caught?

**29.** A package is dropped at time $t = 0$ from a helicopter that is descending steadily at a speed $v_i$. **(a)** What is the speed of the package in terms of $v_i$, $g$, and $t$? **(b)** What vertical distance $d$ is it from the helicopter in terms of $g$ and $t$? **(c)** What are the answers to parts **(a)** and **(b)** if the helicopter is rising steadily at the same speed?

**30.** A certain automobile manufacturer claims that its deluxe sports car will accelerate from rest to a speed of $42.0\ \mathrm{m/s}$ in $8.00\ \mathrm{s}$. **(a)** Determine the average acceleration of the car. **(b)** Assume that the car moves with constant acceleration. Find the distance the car travels in the first $8.00\ \mathrm{s}$. **(c)** What is the speed of the car $10.0\ \mathrm{s}$ after it begins its motion if it can continue to move with the same acceleration?